TP IPv6 Stateful

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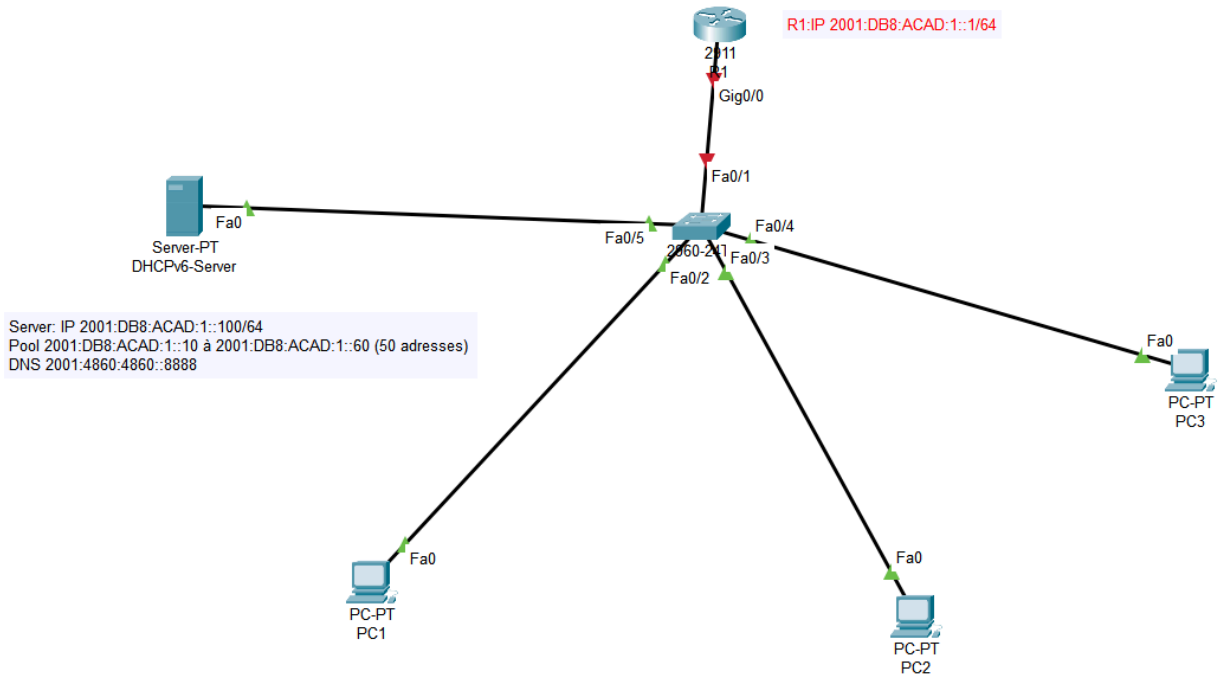
Topologie du réseau



Configuration

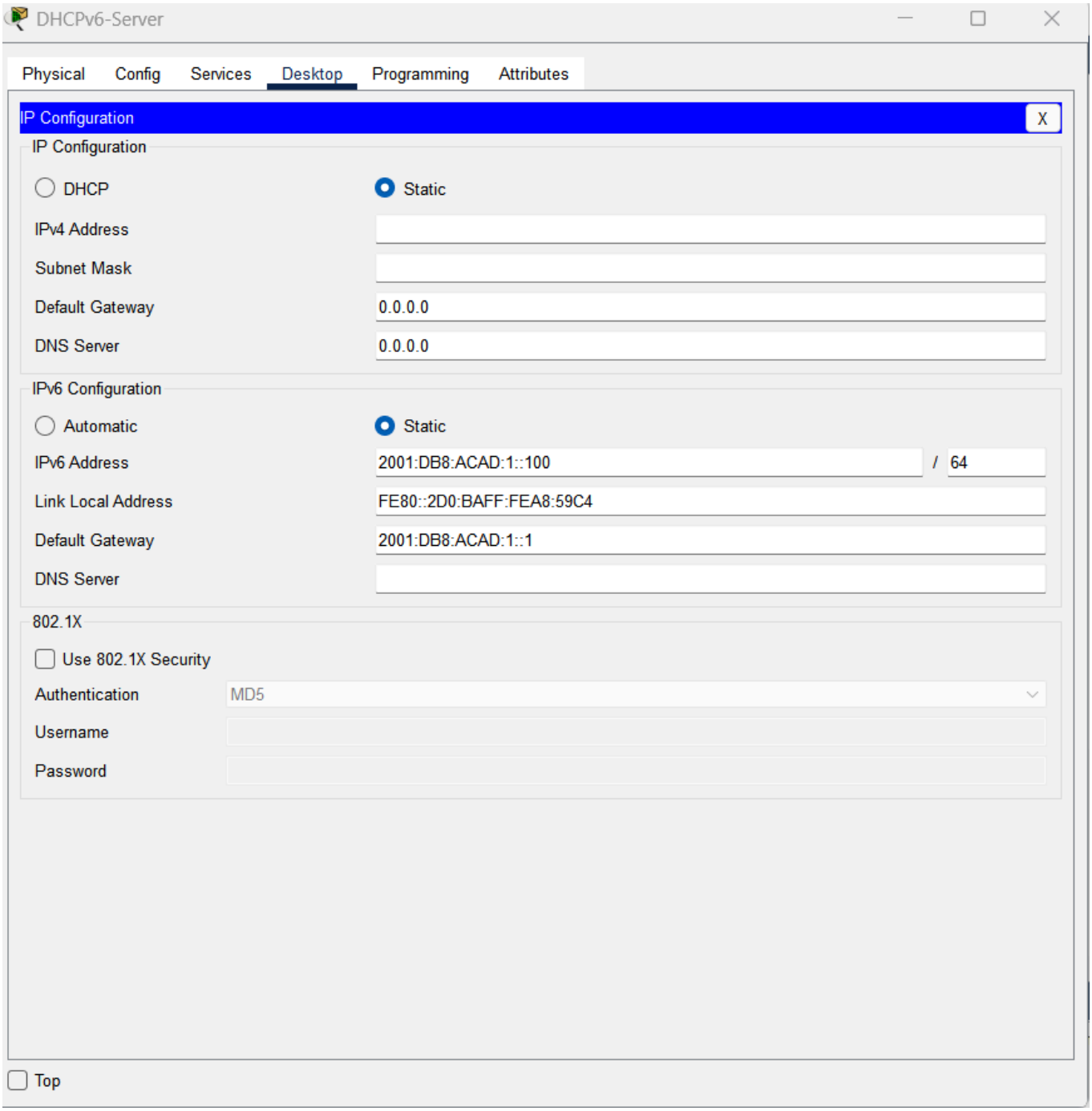
R1

```
R1> enable
R1# configure terminal
R1(config)# no ip domain-lookup
R1(config)# ipv6 unicast-routing
R1(config)# interface GigabitEthernet 0/0
R1(config-if)# description Reseau LAN - Clients DHCPv6 Stateful
R1(config-if)# ipv6 address 2001:DB8:ACAD:1::1/64
R1(config-if)# ipv6 nd managed-config-flag
R1(config-if)# no shutdown
R1(config-if)# exit
R1# show ipv6 interface GigabitEthernet 0/0
```

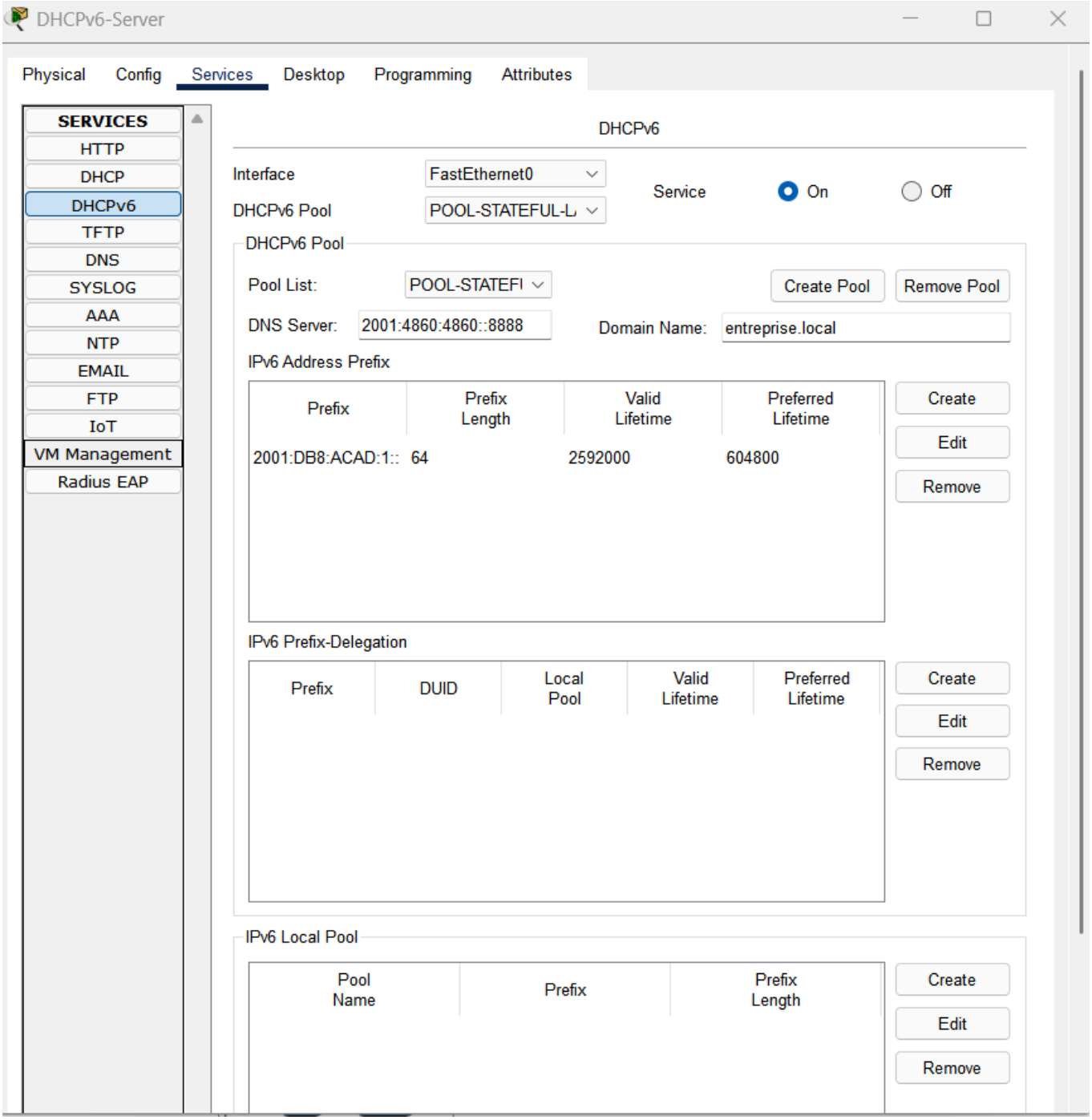
```
R1#show ipv6 interface GigabitEthernet 0/0
GigabitEthernet0/0 is up, line protocol is up
  IPv6 is enabled, link-local address is FE80::2D0:BCFF:FE35:BD01
  No Virtual link-local address(es):
  Global unicast address(es):
    2001:DB8:ACAD:1::1, subnet is 2001:DB8:ACAD:1::/64
  Joined group address(es):
    FF02::1
    FF02::2
    FF02::1:FF00:1
    FF02::1:FF35:BD01
  MTU is 1500 bytes
  ICMP error messages limited to one every 100 milliseconds
  ICMP redirects are enabled
  ICMP unreachables are sent
  ND DAD is enabled, number of DAD attempts: 1
  ND reachable time is 30000 milliseconds
  ND advertised reachable time is 0 (unspecified)
  ND advertised retransmit interval is 0 (unspecified)
  ND router advertisements are sent every 200 seconds
  ND router advertisements live for 1800 seconds
  ND advertised default router preference is Medium
  Hosts use DHCP to obtain routable addresses.
```

DHCPv6-Server

aller dans Desktop -> IPConfiguration

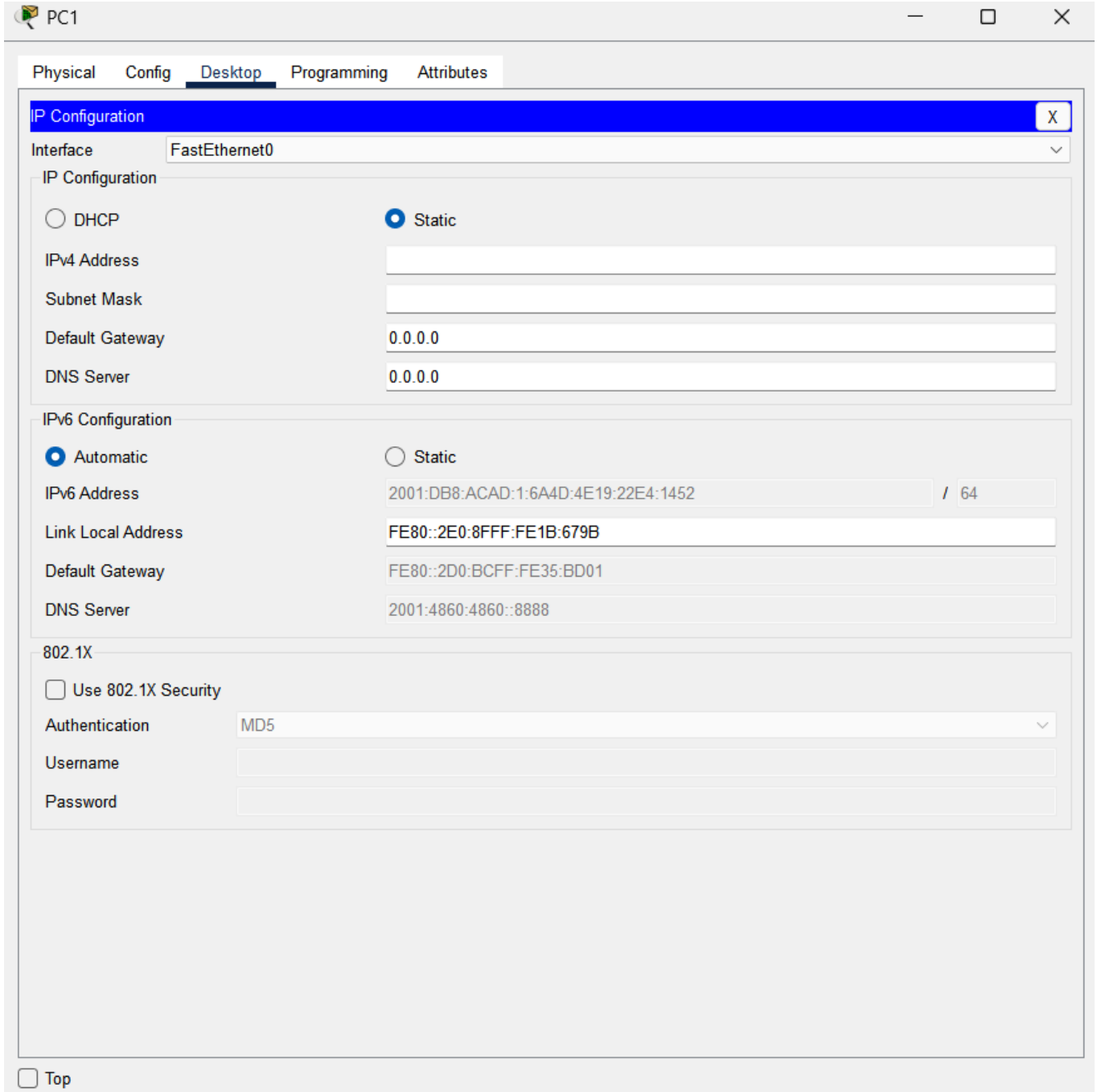


Puis aller dans Services pour activer DHCPv6 pour créer le pool du nom de POOL-STATEFUL-LAN

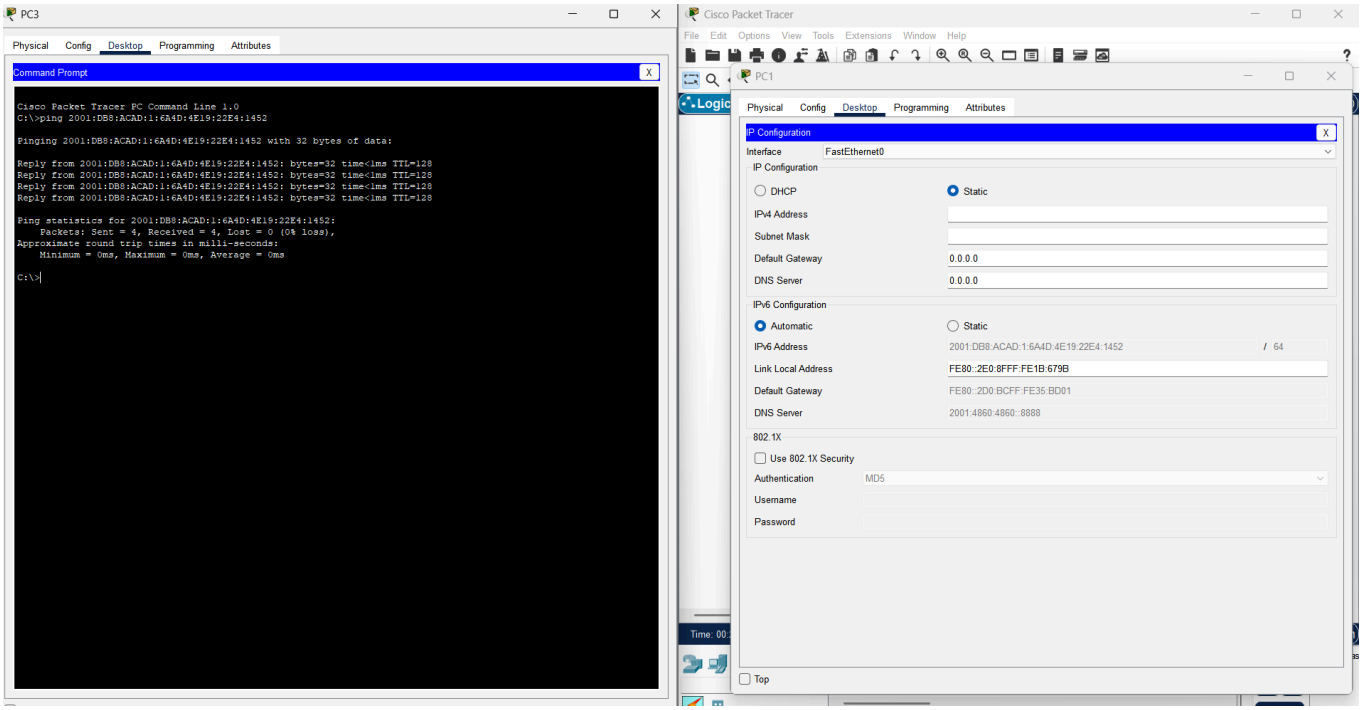

DHCPv6-Server
Physical
Config
Services
Desktop
Programming
Attributes
SERVICES
HTTP
DHCP
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
IoT
VM Management
Radius EAP
DHCPv6
Interface
FastEthernet0
DHCPv6 Pool
POOL-STATEFUL-L
Service
On
Off
DHCPv6 Pool
Pool List:
POOL-STATEFI
Create Pool
Remove Pool
DNS Server:
2001:4860:4860::8888
Domain Name:
entreprise.local
IPv6 Address Prefix
Prefix
Prefix Length
Valid Lifetime
Preferred Lifetime
2001:DB8:ACAD:1::
64
2592000
604800
Create
Edit
Remove
IPv6 Prefix-Delegation
Prefix
DUID
Local Pool
Valid Lifetime
Preferred Lifetime
Create
Edit
Remove
IPv6 Local Pool
Pool Name
Prefix
Prefix Length
Create
Edit
Remove

PC

Basculer tout les PC en attribution automatique d'adresse IP



Vérification avec des ping


PC3
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 2001:DB8:ACAD:1:6A4D:4E19:22E4:1452
Pinging 2001:DB8:ACAD:1:6A4D:4E19:22E4:1452 with 32 bytes of data:
Reply from 2001:DB8:ACAD:1:6A4D:4E19:22E4:1452: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:ACAD:1:6A4D:4E19:22E4:1452: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:ACAD:1:6A4D:4E19:22E4:1452: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:ACAD:1:6A4D:4E19:22E4:1452: bytes=32 time<1ms TTL=128
Ping statistics for 2001:DB8:ACAD:1:6A4D:4E19:22E4:1452:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>
Cisco Packet Tracer
File
Edit
Options
View
Tools
Extensions
Window
Help
PC1
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
Subnet Mask
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
2001:DB8:ACAD:1:6A4D:4E19:22E4:1452
/ 64
Link Local Address
FE80::2E0:8FFF:FE1B:679B
Default Gateway
FE80::2D0:BCFF:FE35:BD01
DNS Server
2001:4860:4860::8888
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top